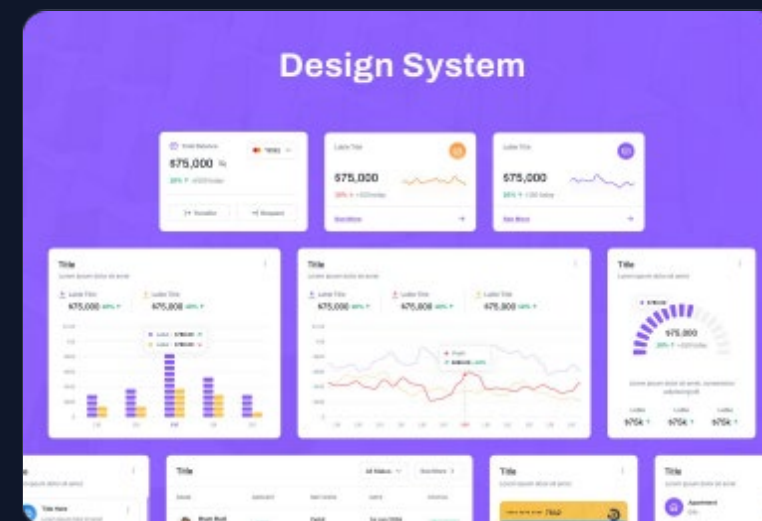


AUTO LOAN SECURITISATION ANALYTICS

Comprehensive Module Framework for Portfolio Performance,
ECL Staging & Cash Flow Waterfall Modeling



CORE SYSTEM MODULES



MODULE 1: DATA MODEL

Star schema architecture.

- Fact performance tables
- Dimension relationships
- High-speed DAX engine



MODULE 2: PORTFOLIO KPI

Balance & characteristic metrics.

- Active vs defaulted balances
- WAC, WAM, WALA calculations
- Weighted LTV & CIBIL



MODULE 3: DELINQUENCY

Early-warning monitoring.

- 30 / 60 / 90+ DPD buckets
- Roll-rate migration matrices
- Early trigger warnings



MODULE 4: IFRS 9 ECL

Staging & impairment engine.

- Stage 1, 2, 3 classification
- $PD \times LGD \times EAD$ formulas
- Provisioning variance



MODULE 5: CASH WATERFALL

Tranche cash allocation.

- Senior / Mezz priority
- OC & IC coverage ratios
- Reserve account drawdowns



MODULE 6: INVESTOR TAPES

Automated rating agency reports.

- Standard ESMA data tapes
- Excel-backed audit checks
- Automated PDF summaries

MODULE 1: DATA ARCHITECTURE & STAR SCHEMA

DIMENSION TABLES

DimLoan: Account IDs, origination date, tenure, interest rates.

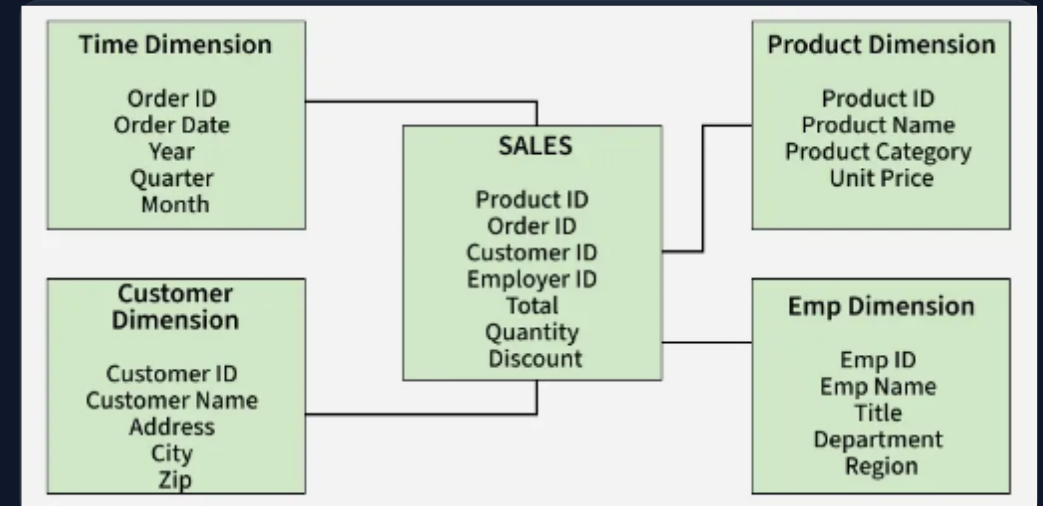
DimBorrower: CIBIL credit score, income band, DTI, employment type.

DimVehicle: New/used tag, LTV ratio, manufacturer class.

FACT TABLES

FactMonthlyPerformance: Monthly collections, principal & interest split.

FactDPDSnapshot: Monthly days-past-due status and roll-rate flags.



MODULE 2: PORTFOLIO ANALYTICS & UNDERWRITING KPIS

A. Balance Metrics

PORTFOLIO OUTSTANDING

- Total active pool balance tracking
- Current vs defaulted balance split
- Principal run-off rates

B. Time Metrics

WAC / WAM / WALA

- Weighted Average Coupon (WAC)
- Weighted Average Maturity (WAM)
- Weighted Average Loan Age (WALA)

C. Credit Quality

RISK CHARACTERISTICS

- Weighted LTV ratio: **78.4%**
- Weighted CIBIL score: **742**
- Weighted DTI ratio: **34.1%**

⚡ **DAX Logic:** Dynamic weighted-average measures recalculate instantaneously upon slicer filtering across originations, geographies, and car segments.

MODULE 3: DELINQUENCY BUCKETS & ROLL-RATE MATRIX

1. DPD BUCKET TRACKING

Categorization into strict delinquency bands:

- Current (0 DPD)
- 1 - 30 DPD (Early Alert)
- 31 - 60 DPD (Moderate Risk)
- 61 - 90 DPD & 90+ DPD (Default)

2. ROLL-RATE MATRIX

Month-on-Month transition probabilities:

- Roll-forward rates (e.g. 30 → 60 DPD)
- Cure rates (e.g. 30 → Current)
- Write-off / Default entry tracking

3. EARLY TRIGGER ALERTS

Automated portfolio health monitors:

- 30+ DPD threshold alerts (> 2.5%)
- Cumulative loss limit breach flags
- Originator servicer warning signals



ANALYTICAL OBJECTIVE

Predict eventual default migration early in the loan lifecycle to initiate targeted collection strategies before Stage 3 impairment.

MODULE 4: IFRS 9 EXPECTED CREDIT LOSS (ECL) ENGINE

PART A

STAGING CRITERIA & LOGIC

- Stage 1 (Performing):** 0-30 DPD. 12-month ECL provision applied.
- Stage 2 (SICR):** 31-90 DPD or Significant Increase in Credit Risk. Lifetime ECL applied.
- Stage 3 (Credit Impaired):** 90+ DPD / Default. Lifetime ECL provision required.

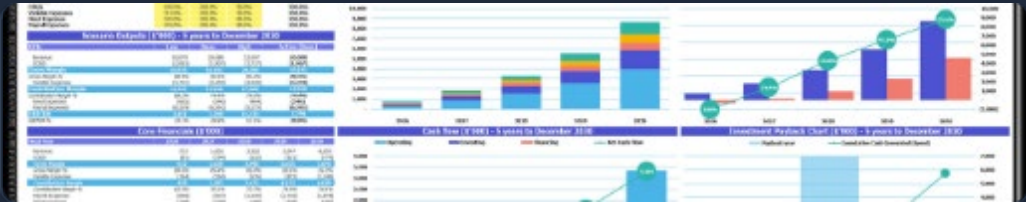
ECL CORE FORMULA

$$\text{ECL} = \text{PD} (\%) \times \text{LGD} (\%) \times \text{EAD} (\text{₹})$$

PART B

ECL MODEL COMPONENTS

- Probability of Default (PD):** Historical transition matrices & forward-looking macro overlays.
- Loss Given Default (LGD):** Vehicle collateral recovery haircut models.
- Exposure at Default (EAD):** Amortization balance plus future drawdowns.



MODULE 5: CASH FLOW WATERFALL & TRANCHE ALLOCATION

PRIORITY OF PAYMENTS EXECUTION ORDER

1. Collections

>

2. Senior Fees

>

3. Class A Note

>

4. Class B Note

>

5. Reserve Top-up

>

6. Equity Residual

OVERCOLLATERALISATION (OC)

Maintains minimum asset-to-debt ratios for rating protection.

INTEREST COVERAGE (IC)

Checks available interest cash flows against note coupons.

RESERVE ACCOUNT MECHANICS

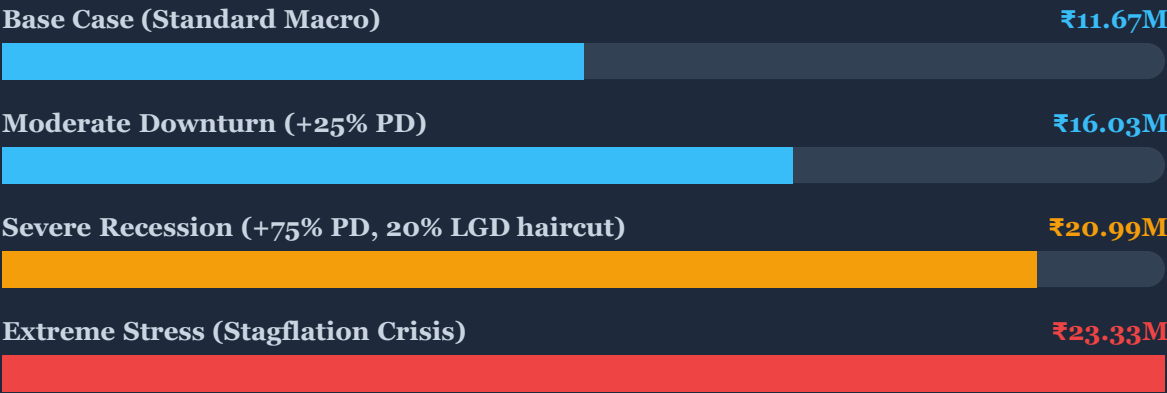
Draws cash during shortfalls & replenishes in excess spread.

MODULE 6: STRESS TESTING & MACRO SCENARIO ENGINE

STRESS PARAMETERS & SHOCKS

- PD Multipliers:** 1.25x - 2.50x default rate increase
- LGD Haircuts:** 10% - 30% reduction in vehicle recovery value
- CPR / SMM Prepayment Speeds:** High/low prepayment shifts

ECL STRESS PROVISION IMPACT (₹ MILLIONS)



MODULES 7 & 8: INVESTOR REPORTING & AUDIT GOVERNANCE



MODULE 7: INVESTOR REPORTING

- Automated Monthly Pool Summary PDF
- Delinquency & Roll-Rate distribution
- Cash Waterfall payout reports
- Trigger breach log notifications



MODULE 8: RATING AGENCY TAPE

- ESMA / CRISIL loan-level data tape
- Pool static performance statistics
- WAC / WAM / WALA automated extract
- EAD & recovery data fields



AUDIT & RECONCILIATION

- 100% Excel model cross-validation
- Completeness & null value integrity checks
- DAX vs SQL data verification
- Row-Level Security (RLS) enforcement

PROJECT DELIVERABLES & VALUE REALIZED

8

Integrated Analytics Modules

80+

DAX & Financial Measures

IFRS 9

ECL Engine & Stress Testing

100%

Automated Data Tape

An end-to-end, production-ready framework providing institutional investors, credit risk managers, and rating agencies with complete transparency into auto loan securitisation portfolios.